

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 6



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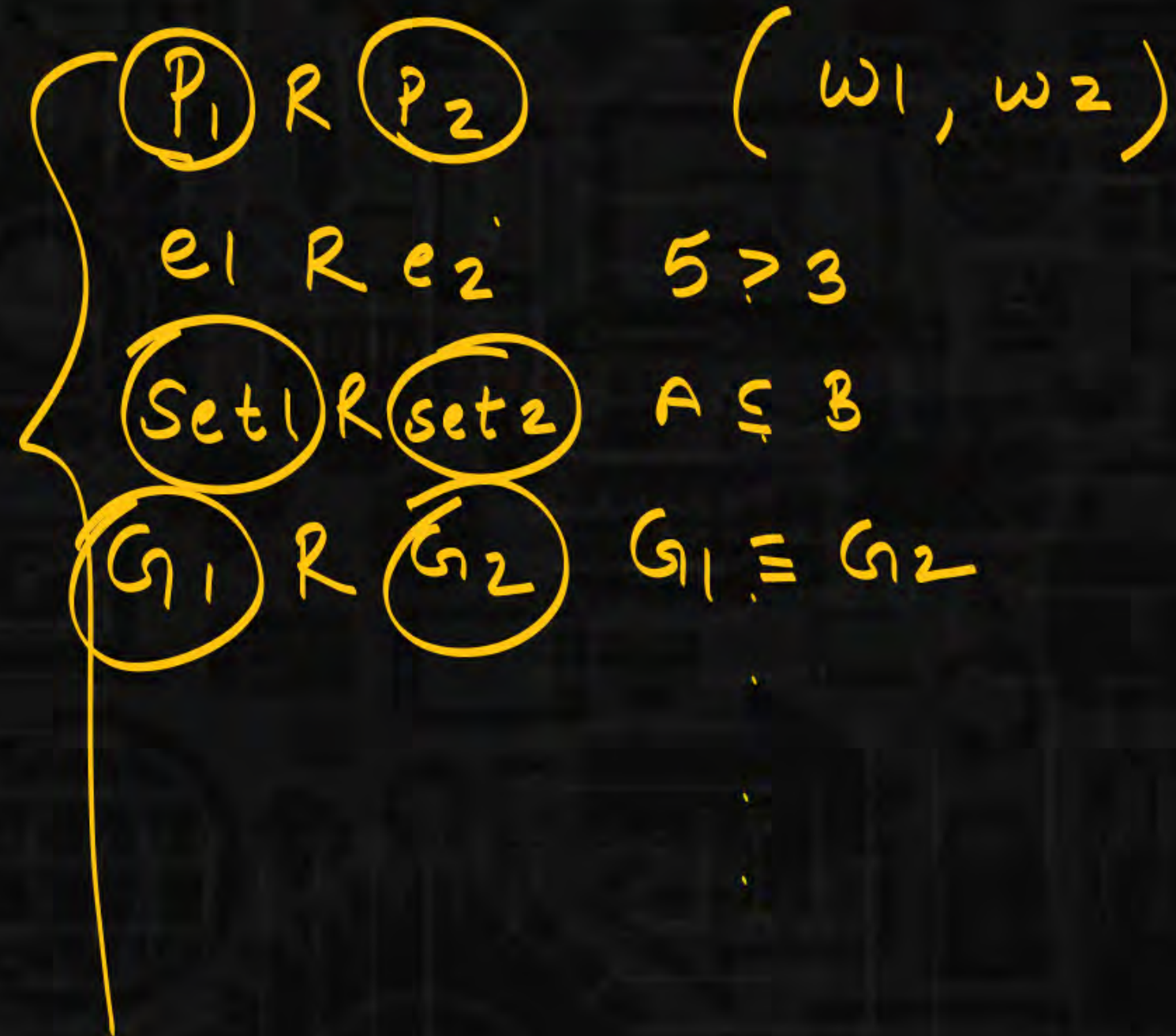
TOPICS TO BE COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

Relations



Set

$$A = \{1, 2, 3\}$$

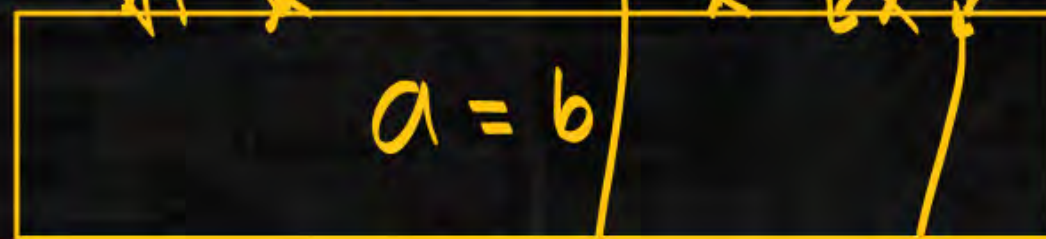
$$B = \{2, 3\}$$

Relations



$$A = \{1, 2, 3\} \quad B = \{2, 3\}$$

$$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3), (3, 2), (3, 3)\}$$



$$R_1 = \{(2, 2), (3, 3)\} \checkmark$$

Relations



Relation: subsets of cross product of 2 set's A, B.

$$|A|=m \quad |B|=n. \quad |A \times B| = \underline{m \cdot n}.$$

Total no. of
Relations = $2^{m \cdot n}$

eg: no. of Relations = 2^{n^2}

$$|A|=n. \quad |A \times A| = n^2$$

Relations



$$A = \{1, 2, 3\} \quad B = \{2, 3\}$$

$$A \times B = \left\{ \begin{array}{l} (1, 2) \quad (1, 3) \\ (2, 2) \quad (2, 3) \\ (3, 2) \quad (3, 3) \end{array} \right\}$$

$$A \times B \rightarrow \boxed{a < b} \rightarrow R_3 = \left\{ \begin{array}{l} (1, 2) \\ (1, 3) \\ (2, 3) \end{array} \right\}$$

$$R_1 = \{(a, b) \mid a = b\}$$

$$A \times B \rightarrow \boxed{a = b} \rightarrow \left\{ (2, 2) \quad (3, 3) \right\}$$

$$A \times B \rightarrow \boxed{a > b} \rightarrow R_2 = \left\{ (3, 2) \right\}$$

$$\underline{R_2 = \{(a, b) \mid a > b\}}$$

Relations

$$Z = \{ \dots \} \quad Z = \{ \dots \} \quad R_1 = \{ (a, b) \mid a = b \}$$

$Z \times Z$

$(0, 0), (1, 1), (1, 2), (1, 3)$

$(2, 1), (2, 2)$

$a = b$

$a \geq b$

$R_2 = \{ (a, b) \mid a \geq b \}$

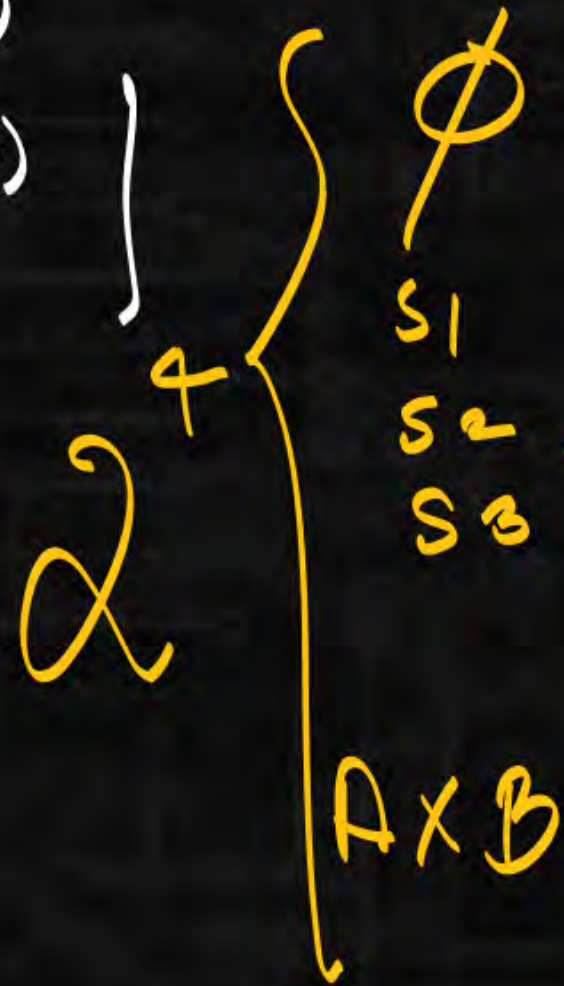
Relations



$$A = \{1, 2\} \quad B = \{2, 3\}$$

$$A \times B = \left\{ \begin{array}{l} (1, 2) \quad (1, 3) \\ (2, 2) \quad (2, 3) \end{array} \right\}$$

$$|A \times B| = 4$$



$$A \times B \rightarrow \boxed{a+b=1} \rightarrow R_1 = \emptyset$$

$$a+b=3 \quad \boxed{R_2 = \{(1, 2)\}}$$

$$a+b=4 \quad R_3 = \left\{ \begin{array}{l} (1, 3) \\ (2, 2) \end{array} \right\}$$

$$a+b=5 \quad R_4 = \{(2, 3)\}$$

$$a+b \leq 100 \quad R_5 = \{ \dots \}$$

Relations

$A \rightarrow$ non empty set.

$$A \times A = \{ \dots \}$$

Symmetric Relation:

$$\forall a \forall b [(a, b) \in R \longrightarrow (b, a) \in R]$$

$$R_1 = \{ \overset{a, b}{(1, 2)} \}$$

not symmetric.

$$(a, b) \in R \longrightarrow (b, a) \in R.$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ (1, 2) \in R & \longrightarrow & (2, 1) \in R \\ \text{T} & \longrightarrow & \text{F} \end{array}$$

$$a=1 \quad b=2$$

Relations

$$A = \{1, 2, 3\}$$

$$A \times A = \{ \dots \}$$

$$\left\{ \underline{(a, b) \in R} \rightarrow \underline{(b, a) \in R} \right\}$$

Symmetric:

$$R_2 = \{ (\underline{1, 2}), (2, 1) \}$$

$$(a, b) \in R \rightarrow (b, a) \in R.$$

$$\underline{(1, 2) \in R} \rightarrow \underline{(2, 1) \in R}.$$

T T

it is symmetric

$$R_3 = \{ \dots \}$$

Relation

$$\underline{(a, b) \in R} \rightarrow (b, a) \in R.$$

f →

Relations



$$R_3 = \{ (1, 1) (2, 2) \} \text{ Symmetric Relation.}$$

$$(a, b) \in R \rightarrow (b, a) \in R.$$

$$(1, 1) \in R \rightarrow (1, 1) \in R.$$

$$a=1 \quad b=1.$$

$$R_1 = \{ \} \checkmark$$

$$R_2 = \{ (1, 2) \} \times$$

$$R_3 = \{ (1, 2) (2, 1) \} \checkmark$$

$$R_4 = \{ (1, 1) \}$$

Relations



$$R_1 = \{ \} \checkmark$$

$$R_2 = \{ (1, 3) (3, 1) \} \checkmark$$

$$R_3 = \{ (1, 2) (2, 1) (1, 1) \} \checkmark$$

$$R_4 = \{ (2, 3) (2, 2) \} \checkmark$$

$$(a, b) \in R \rightarrow (b, a) \in R.$$

$$\frac{(2, 3) \in R}{T} \rightarrow \frac{(3, 2) \in R}{F}$$

note:

- Symmetric demands flipping.
- no problem with same element

Relations



$$R_1 = \left\{ \underbrace{(1,1)}_{e_1} \underbrace{(1,2)}_{e_2} \underbrace{(2,1)}_{e_3} \underbrace{(2,3)}_{e_4} \underbrace{(3,2)}_{e_5} \right\}$$

not symmetric.

$$R_1 = \left\{ \overset{\times}{(1,2)} \overset{\checkmark}{(1,1)} \overset{\checkmark}{(2,3)} \overset{\checkmark}{(3,2)} \right\}$$
$$\left\{ \overset{\checkmark}{(1,1)} \overset{\checkmark}{(1,2)} \overset{\checkmark}{(2,3)} \overset{\checkmark}{(2,1)} \overset{\checkmark}{(3,2)} \right\} \checkmark$$

Relations



$$R_1 = \left\{ \underbrace{(1,1)}_P, \underbrace{(1,2)(2,1)}_P \right\}$$

how many symmetric Relations
are possible?

$$R_2 = \left\{ \underbrace{(1,2)(2,1)}_P \right\} \checkmark$$

$$R_3 = \left\{ \underbrace{(1,1)}_P \right\} \checkmark$$

Relations



$$n = 3$$

$$\text{boxes} = n^2 - n / 2$$

$$A = \{1, 2, 3\}$$

$$A \times A = \left\{ \begin{array}{l} \boxed{(1,1)} \quad \boxed{(2,2)} \quad \boxed{(3,3)} \quad \boxed{\overset{h}{(1,2)} \overset{w}{(2,1)}} \quad \boxed{(1,3) (3,1)} \quad \boxed{(2,3) (3,2)} \\ \downarrow \quad \uparrow \quad \uparrow \quad \downarrow \quad \uparrow \quad \uparrow \\ \text{2 choices} \quad 2 \quad \text{2 choices} \quad \text{2 choices} \quad \text{2 choices} \quad \text{2 choices} \end{array} \right\}$$



Relations



$$A = n$$
$$A = \{1, 2, 3\} \quad A \times A = n^2 \quad A \times A = \{$$

$$n = 3$$

Diagonal elements = $3 = n$.

non diagonal elements
 $= n^2 - n$.

$$\text{boxes} = \frac{n^2 - n}{2}$$



}

Relations



Diagonal bones:

$$2^n \cdot 2^{\frac{n^2-n}{2}}$$

of symmetric Relations = $2^n \cdot 2^{\frac{n^2-n}{2}}$
 OR
 $2^{n(n+1)/2}$

$$= 2^n \cdot 2^{\frac{n^2-n}{2}}$$

$$n + \frac{n^2-n}{2} = \frac{2n + n^2 - n}{2}$$

$$2^{\frac{2n + n^2 - n}{2}} = 2^{\frac{n^2 + n}{2}}$$

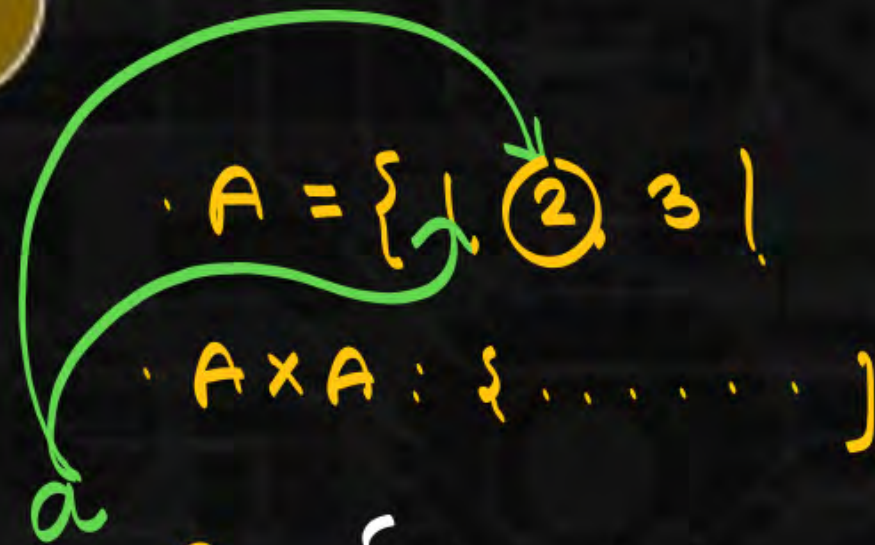
$$= 2^{\frac{n^2+n}{2}} = 2^{\frac{n(n+1)}{2}}$$

Relations

$A \rightarrow$ non empty set

Reflexive:

$$\underline{\forall a \in A (a, a) \in R.}$$



$$R_1 = \left\{ \underbrace{(1,1)}_{PV}, (1,2), (2,1), (2,3) \right\} \times$$

Relations

$$A = \{1, 2, 3\}$$

$$\forall a \in A (a, a) \in R$$

$$R_2 = \{ (2, 1) (2, 3) \underline{(2, 2)} (1, 3) \underline{(1, 1)} (3, 1) \underline{(3, 3)} \} \quad \text{Reflexive}$$

$$R_3 = \{ \quad \} \quad \text{not Reflexive}$$

$$R_4 = \{ (1, 1) (2, 2) \}$$

$$A = \{1, 2, 3\}$$

Relations

$$R_1 = \left\{ \boxed{11 \quad 22 \quad 33} \right\} \checkmark$$

$$R_3 = \left\{ (11) (22) (33) (12) (21) \right\} \checkmark$$

$$R_2 = \left\{ \boxed{11 \quad 22 \quad 33} (12) \right\} \checkmark$$

Relations

$$A = \{1, 2, 3\} \quad A \times A =$$

$$|A| = n = 3.$$

Diagonal elements = n .

non diagonal elements = $n^2 - n$.

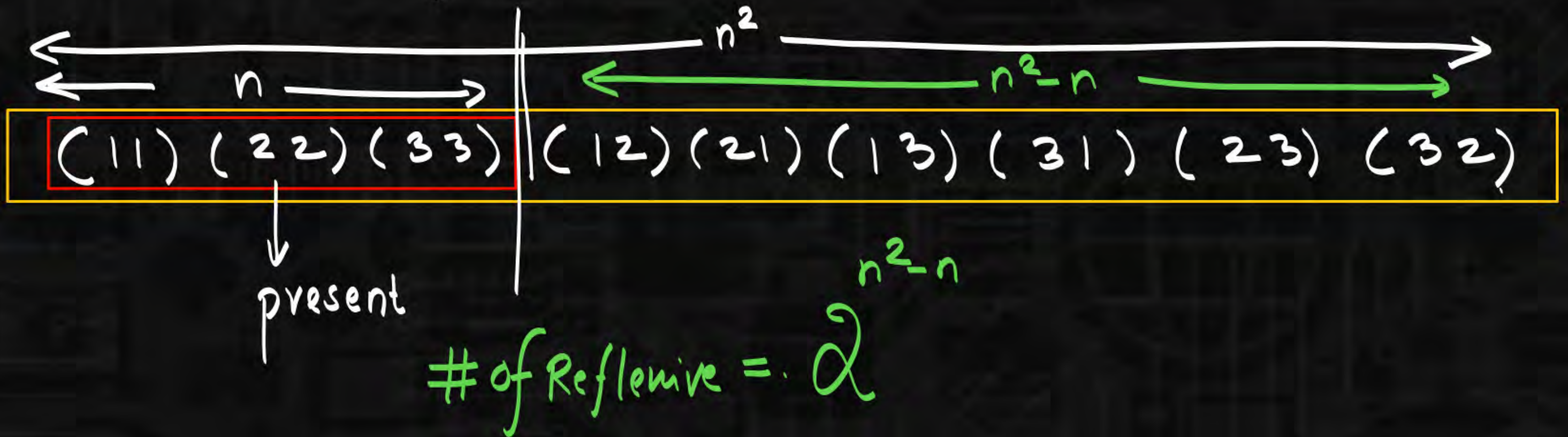
$$\begin{matrix} n & n^2 - n \\ 1 & 2 \end{matrix}$$

(11)	(12)	(13)
(21)	(22)	(23)
(31)	(32)	(33)

→ present (1 choice)

Relations

Total no. of reflexive Relations = 2^{n^2-n}



Relations

$$|A| = n$$

$$|A \times A| = n^2$$

$$\# \text{ Symmetric} = 2^n \cdot 2^{n^2-n/2}$$

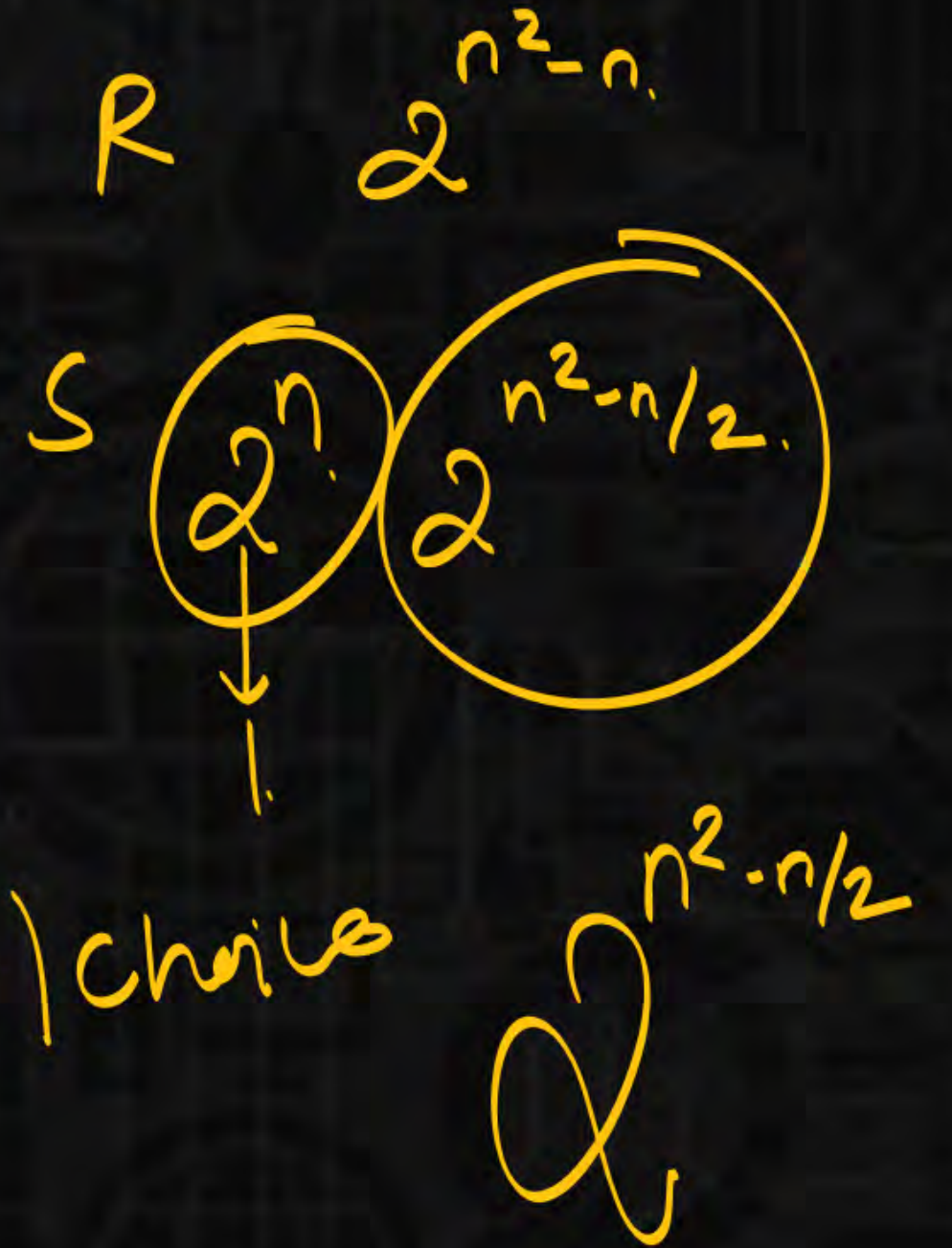
$$\# \text{ Reflexive} = 2^{n^2-n}$$

$$\# (R \cap S) =$$

Relations



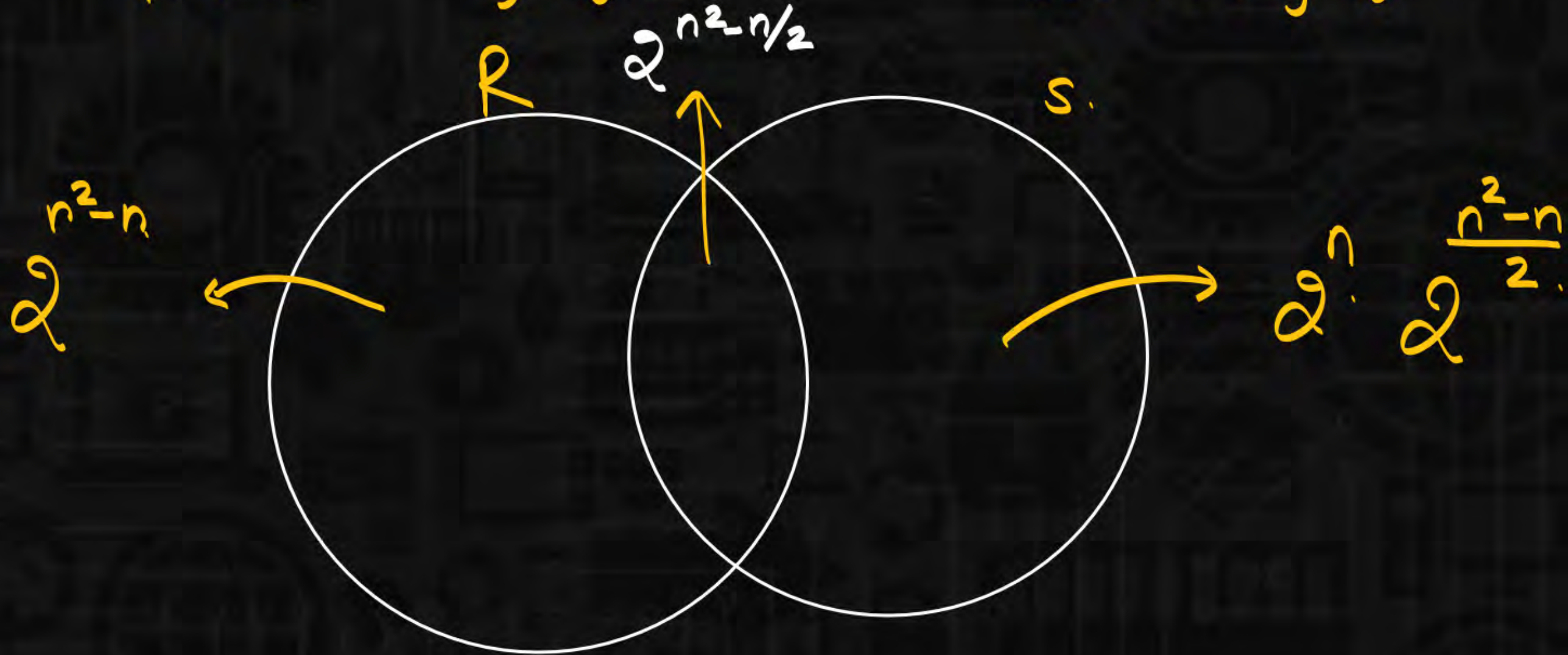
11	12	13
21	22	23
31	32	33



Relations

$R \rightarrow$ Total no. of Reflexive

$S \rightarrow$ total no. of symmetric.



Relations

$$R \cup S = R + S - R \cap S.$$

$$= 2^{n^2-n} + 2^n \cdot 2^{\frac{n^2-n}{2}} - 2^{\frac{n^2-n}{2}}.$$

Relations



$$R_1 = \{ (a, b) \mid a + b \leq 3 \}$$

Symm ✓

Symmetric. $(a, b) \in R \longrightarrow (b, a) \in R$ (finite)
or
 $a R b \longrightarrow b R a$.

$$a R b \longrightarrow b R a$$

$$\underline{a + b \leq 3} \longrightarrow \underline{b + a \leq 3}.$$

$$(1, 2) \in R \longrightarrow (2, 1) \in R.$$

$$1 + 2 \leq 3 \longrightarrow 2 + 1 \leq 3$$

Relations

$$R_2 = \{ (a, b) \mid a \leq b \}$$

$$\underline{a R b \rightarrow b R a}$$

$$a \leq b \rightarrow b \leq a$$

Reflexive: $(a, a) \in R$
OR
 $a R a$

$$R_2 = \{ (a, b) \mid a \leq b \} \checkmark$$

$$a \leq a \text{ (for all)}$$

Relations



$$R_1 = \{ (a, b) \mid a = b + 1 \}$$

$$R_3 = \{ (a, b) \mid a + b \leq 0 \}$$

$$R_2 = \{ (a, b) \mid \gcd(a, b) = 1 \}$$

